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Opening the Black Box

Mechanistic Interpretability for AI Agent Tool Selection Using Sparse Autoencoders

Hannes Hapke with **David Cardozo** · 575 Lab, Dataiku Inc.

ABOUT THE SPEAKER

Hi, I'm Hannes Hapke

Director, Engineering at Dataiku, 575 Lab

Google Developer Expert -- Machine Learning

Focus areas:

- Open Source Machine Learning
- Production ML systems & pipelines
- Generative AI and agentic systems

Co-author of four O'Reilly & Manning books, including:

- **Generative AI Design Patterns** (2025)
- **Machine Learning Production Systems** (2024)
- **Building Machine Learning Pipelines** (2020)
- **NLP in Action (1st edition)** (2019)

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Dataiku's Open Source Office



Advancing Responsible AI through open source — production-ready tools that give enterprises **transparency, control, and governance** to deploy AI with confidence.

Our Responsible AI work:

- **Trust** — no black boxes, no hidden behaviour
- **Control** — no vendor lock-in, no opaque dependencies
- **Accountability** — auditable, reproducible, community-reviewed

SIGNATURE PROJECTS

- **Kiji Inspector** — mechanistic interpretability for agent tool selection (*this talk*)
- **Kiji Privacy Proxy** — detect & mask sensitive data before LLM API calls and unmask responses

ALSO CONTRIBUTING TO

vLLM · scikit-learn (sponsor) · Cardinal (active learning) · CodeMirror

dataiku.com/open-source

The Problem: Opaque Agent Decisions

AI agents autonomously select tools (databases, web search, code execution, ...) based on natural language requests.

But **why**?

"Find information about our company's API rate limits."

→ Internal docs search? Or public web search?

Current approaches **fail** to provide mechanistic insight:

1. **Prompt engineering** -- reveals correlations, not causal mechanisms
2. **Behavioral testing** -- characterizes inputs/outputs, not internals
3. **Chain-of-thought** -- plausible narratives $\neq$ true computation

We need to look **inside the model**.

PART I

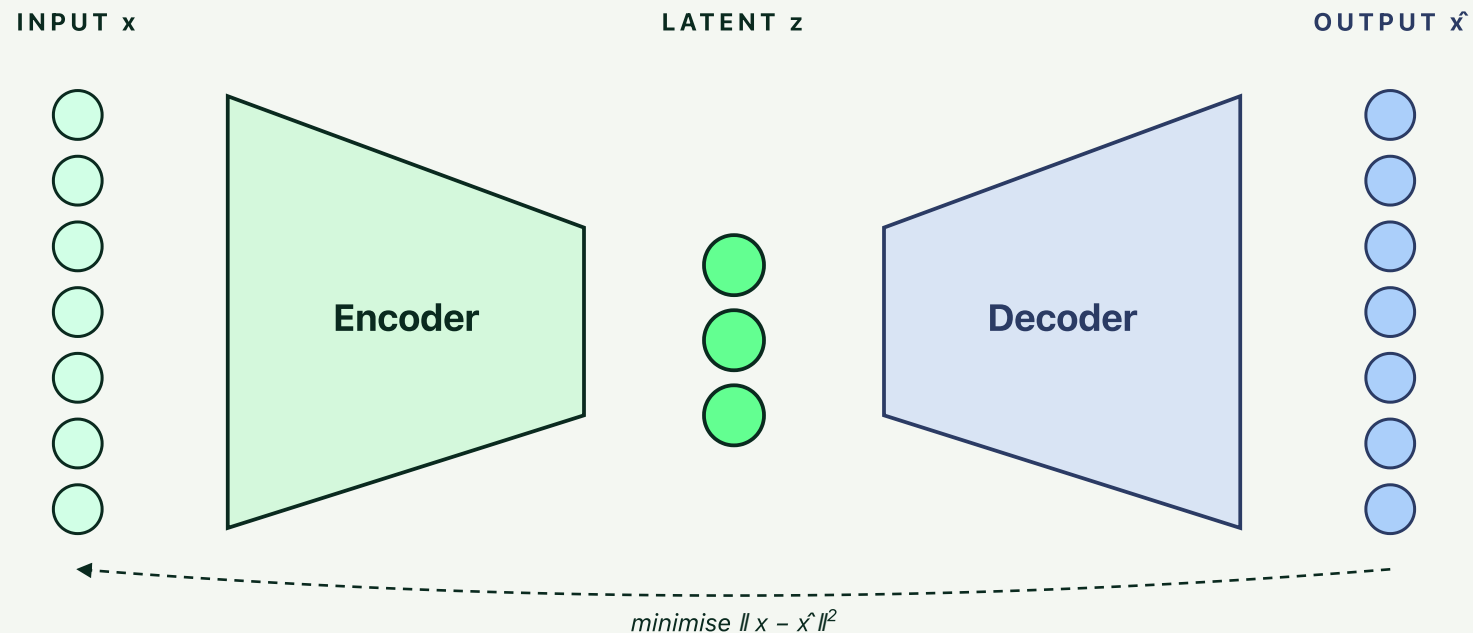
The Solution

LOOKING INSIDE WITH SPARSE AUTOENCODERS

How to Understand Models — Step 1

AUTOENCODERS

A **self-supervised** technique that learns new representations: compress an input through a bottleneck, then reconstruct it. What survives the bottleneck is what the model considers essential.

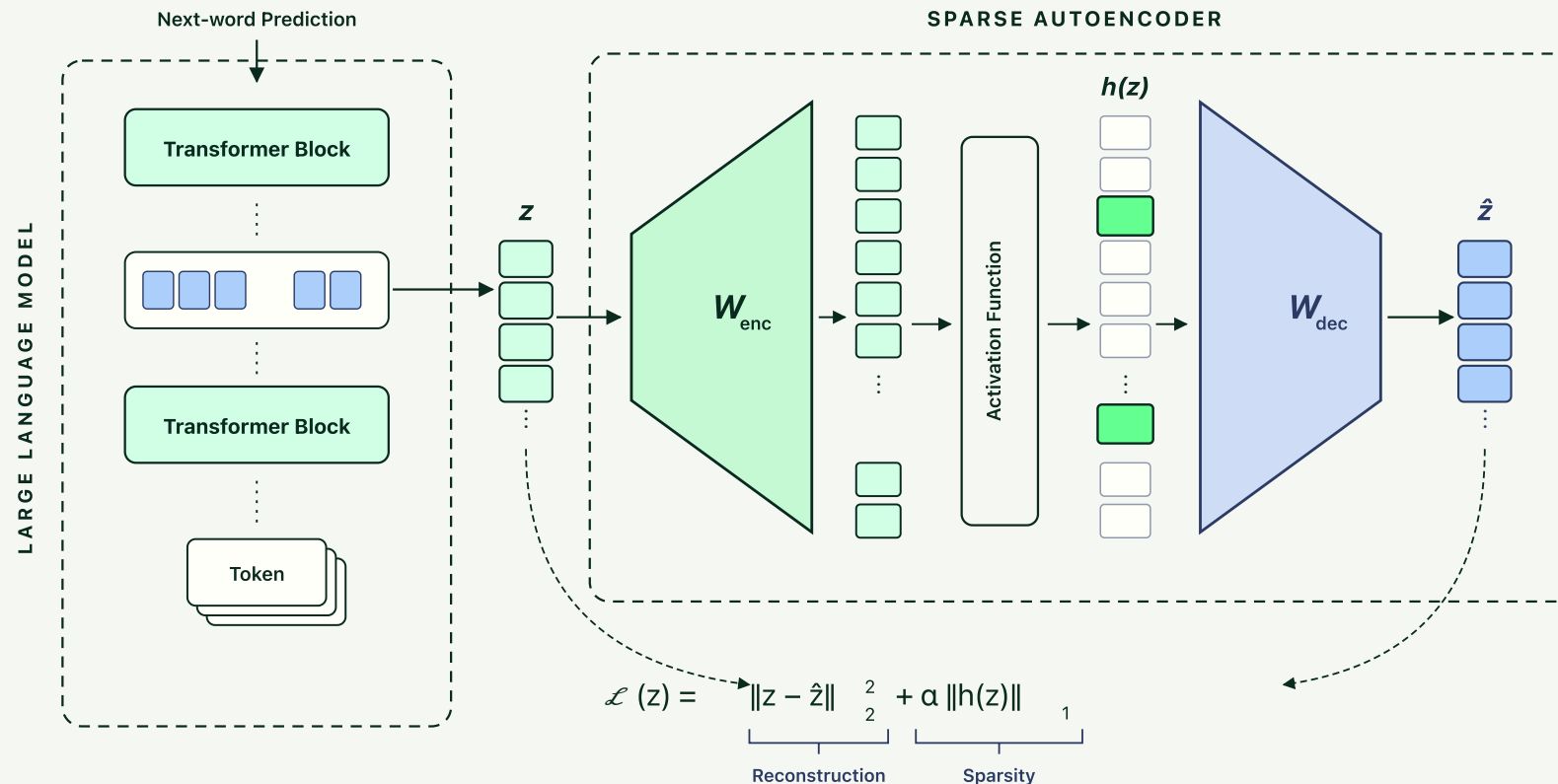


- **Well-understood** -- decades of theory and practice
- **Lossy by design** -- the bottleneck forces the model to *prioritise*

How to Understand Models — Step 2

SPARSE AUTOENCODERS (SAE)

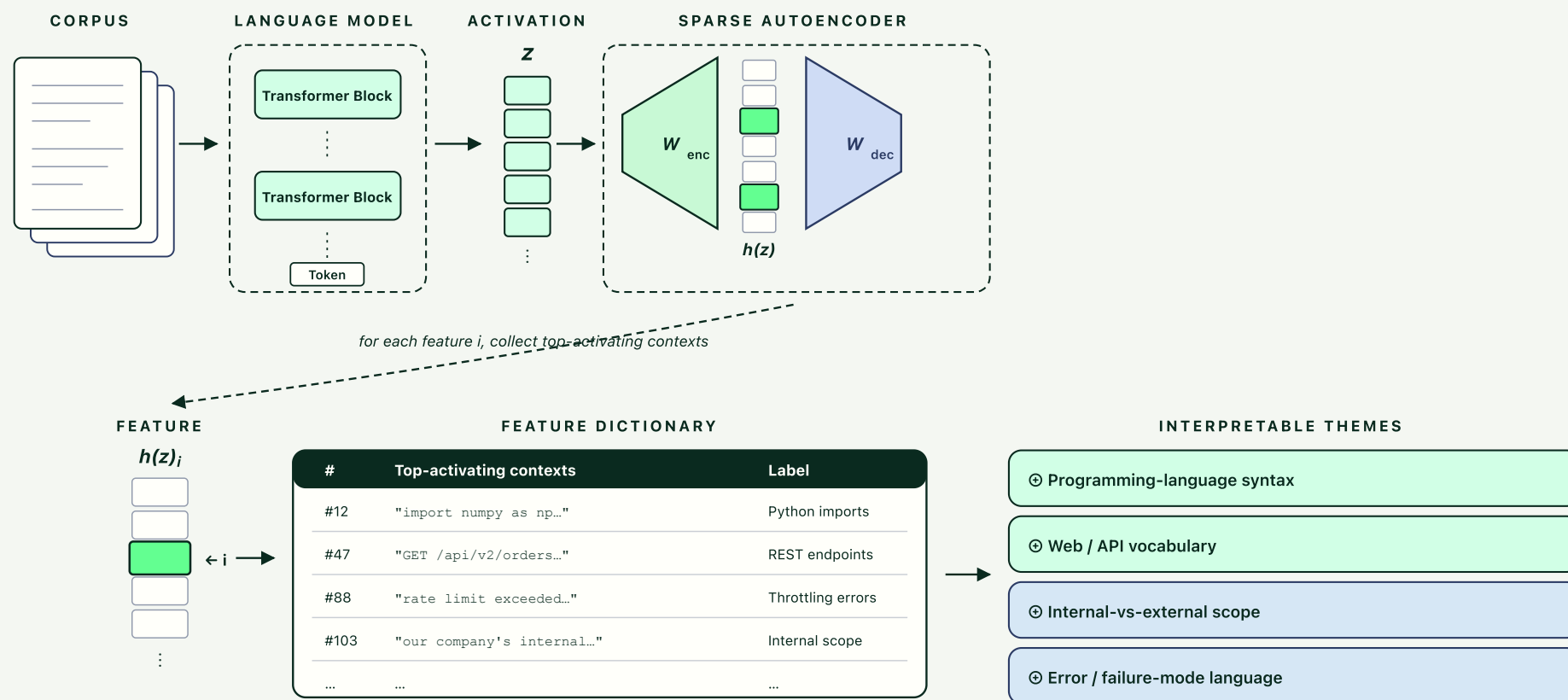
Flip the autoencoder on its head: instead of compressing, **expand** the latent space — but force fewer than **5%** of dimensions to fire on any given input.



How to Understand Models — Step 3

INTERPRETING WHAT THE FEATURES MEAN

A trained feature is just an index. To *label* it, collect the contexts where it fires most strongly — then let an LLM describe the pattern.



What's Novel?

01

DECISION-TOKEN EXTRACTION

Capture activations at the *precise moment* of tool commitment — not averaged over the prompt.

02

PAIRS AS POST-HOC PROBES

The SAE learns the model's natural vocabulary **unsupervised**. Contrastive pairs are statistical probes only — never training signal.

03

TOKEN-LEVEL FUZZING EVALUATION

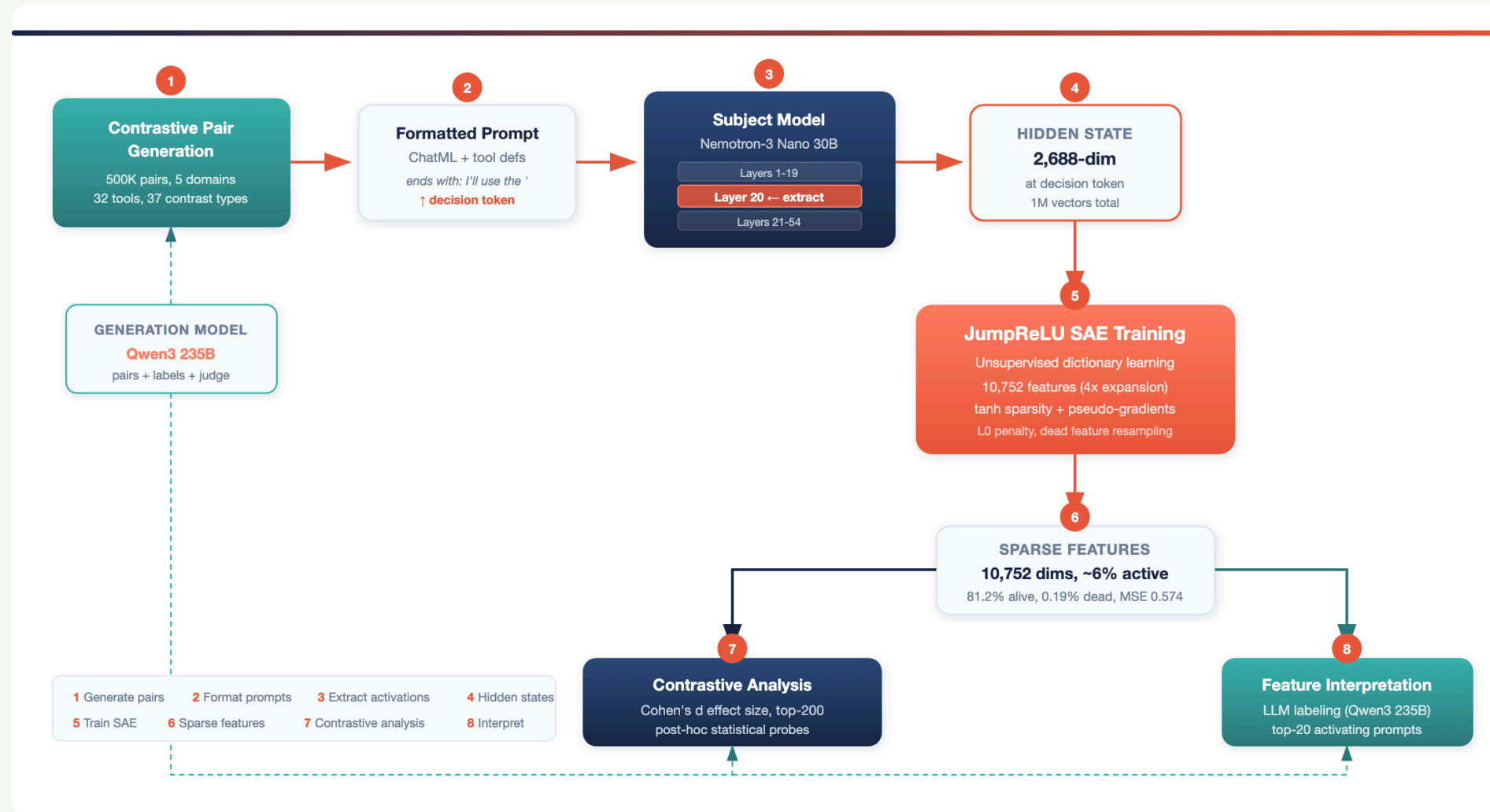
Adapted from Eleuther AI's autointerp — catches labels that are *"right for the wrong reasons."*

04

CAUSAL VALIDATION VIA ABLATION

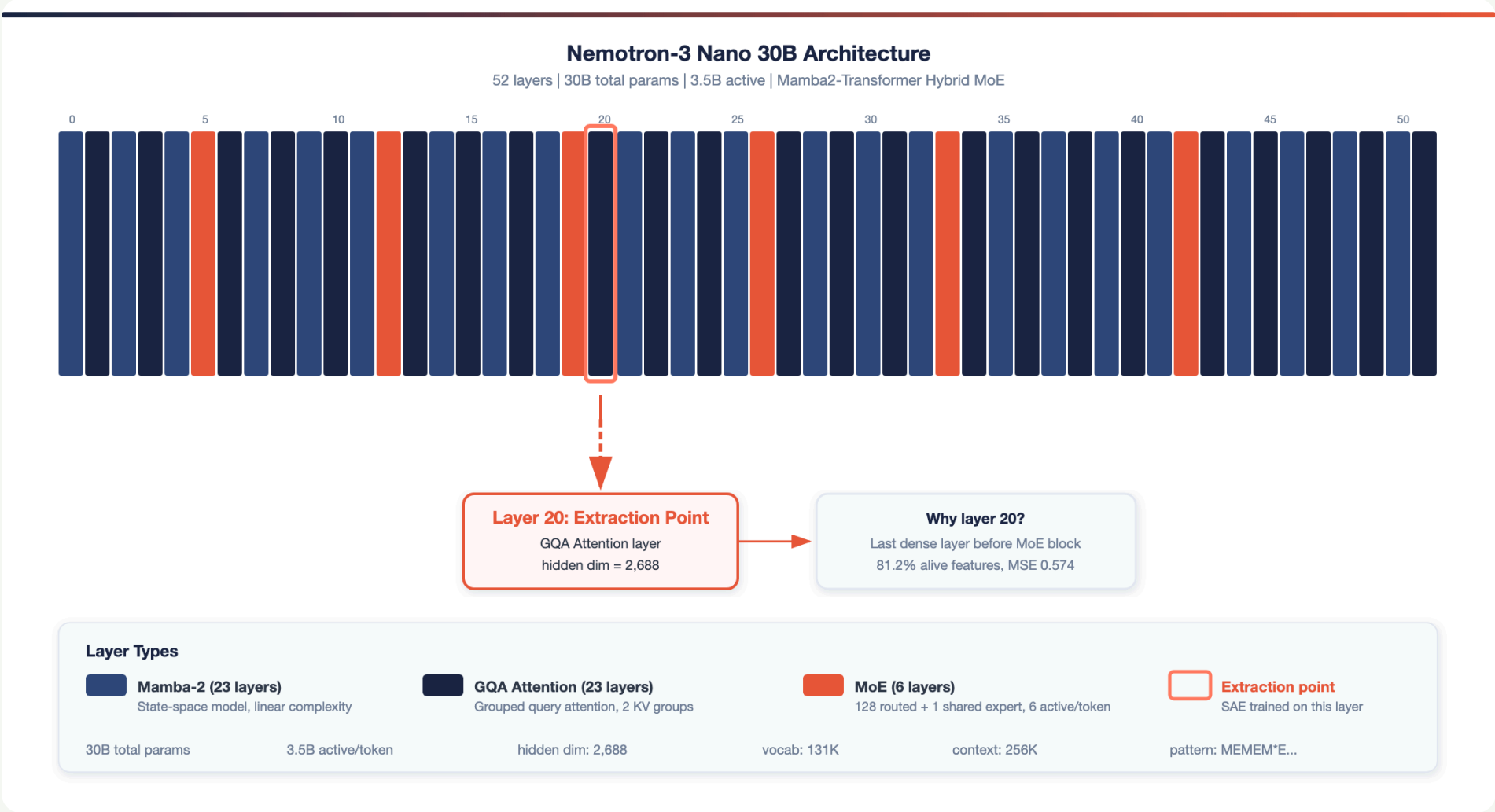
Zero a feature, measure the prediction flip — turns correlations into evidence.

Our Complete Training Pipeline



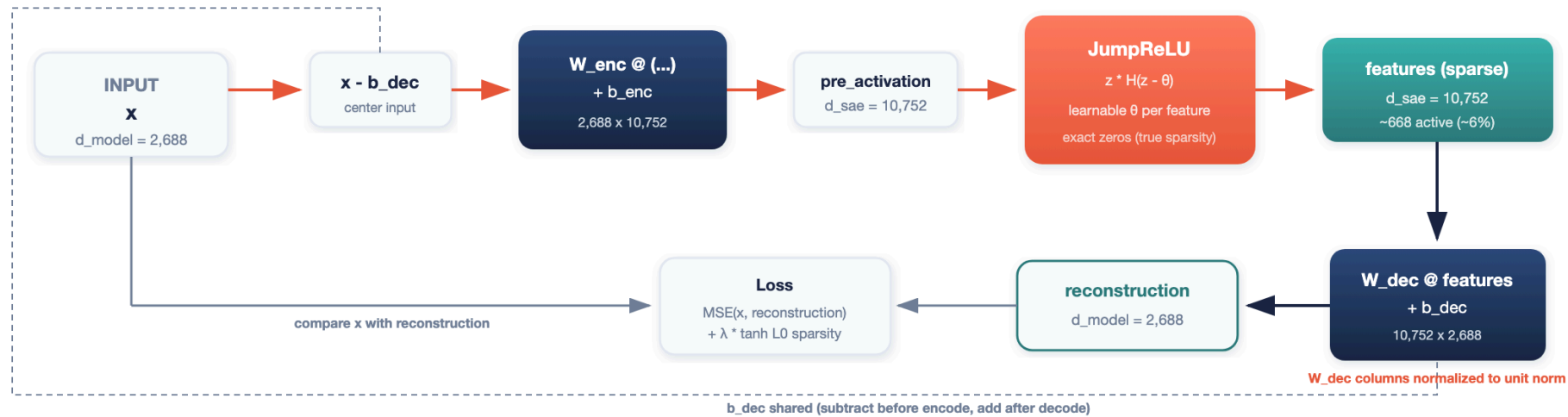
The SAE is trained unsupervised on the activations; contrastive pairs serve only as post-hoc statistical probes.

Nemotron-3 Nano Architecture



Hybrid Mamba2-Transformer MoE, 52 layers, open weights &mdash. We extract activations at **layer 20** (GQA attention).

PyTorch SAE Model Architecture



Parameters

W_{enc} ($2,688 \times 10,752$) **W_{dec}** ($10,752 \times 2,688$) **θ** ($10,752$) thresholds **b_{enc}** ($10,752$) **b_{dec}** ($2,688$) shared
Total: $\sim 58\text{M}$ trainable parameters Activation: JumpReLU with STE pseudo-gradients Loss: $\text{MSE} + \lambda * \tanh\text{-smoothed L0}$

Encoder projects 2,688-dim input to 10,752 sparse features via JumpReLU with learnable per-feature thresholds.

Decision Token Extraction

Every formatted prompt ends with:

```
<|assistant|> I'll use the '
```

The hidden state at this final token is the **decision token** -- the model's internal state at the moment it commits to a tool name.

- Activations extracted at **layer 20** of Nemotron-3-Nano-30B (54-layer MoE) — *layer choice justified below*
- Hidden dimension: **2,688**
- Batched extraction with left-padding for alignment
- Dataset: **1,000,000** activation vectors (500K contrastive pairs)

Contrastive Pair Design

Pairs share the same *intent* but require *different tools*:

Shared Intent	Anchor (tool A)	Contrast (tool B)
Resolve password issue	"How do I reset my password?" → <code>knowledge_base</code>	"I tried resetting 3 times but the email never arrives" → <code>ticket_lookup</code>
Evaluate energy stocks	"Which companies invest in renewables?" → <code>financial_analysis</code>	"Which stocks trade below book value?" → <code>market_data_lookup</code>
Check product version	"What is the latest version?" → <code>file_read</code>	"Set the version to v3.2.1" → <code>file_write</code>

5 domains, 32 tools, 37 contrast types.

JumpReLU SAE Architecture

JumpReLU = ReLU with a learnable per-feature threshold — gives exact zeros but stays trainable via tanh pseudo-gradients (Rajamanoharan et al., 2024).

Encoder:

$$f_i(\mathbf{x}) = \pi_i(\mathbf{x}) \cdot H(\pi_i(\mathbf{x}) - \theta_i)$$

where $\pi_i(\mathbf{x}) = [W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{dec}}) + \mathbf{b}_{\text{enc}}]_i$ and θ_i is a learnable threshold.

Key properties:

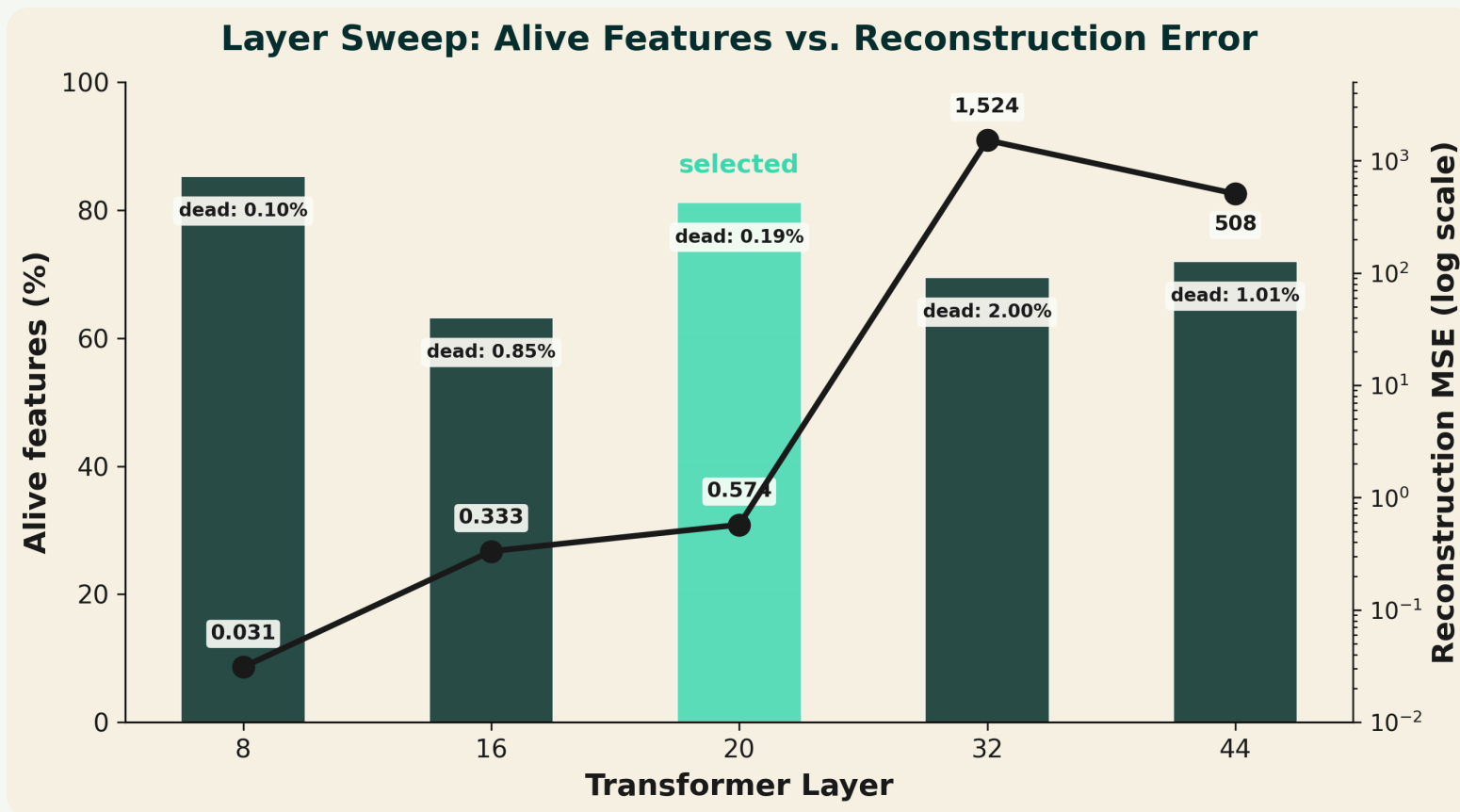
- **Exact sparsity** -- Heaviside step function H produces true zeros
- **Smooth training** -- tanh approximation for gradient flow:

$$\hat{\mathcal{L}}_{\text{sparse}} = \sum_{i=1}^M \text{ReLU} \left(\tanh \left(\frac{\pi_i - \theta_i}{\varepsilon} \right) \right)$$

- **Pseudo-gradients** via rectangular kernel density estimator

Dictionary size $M = 10,752$ ($4 \times$ hidden dim).

Why Layer 20?



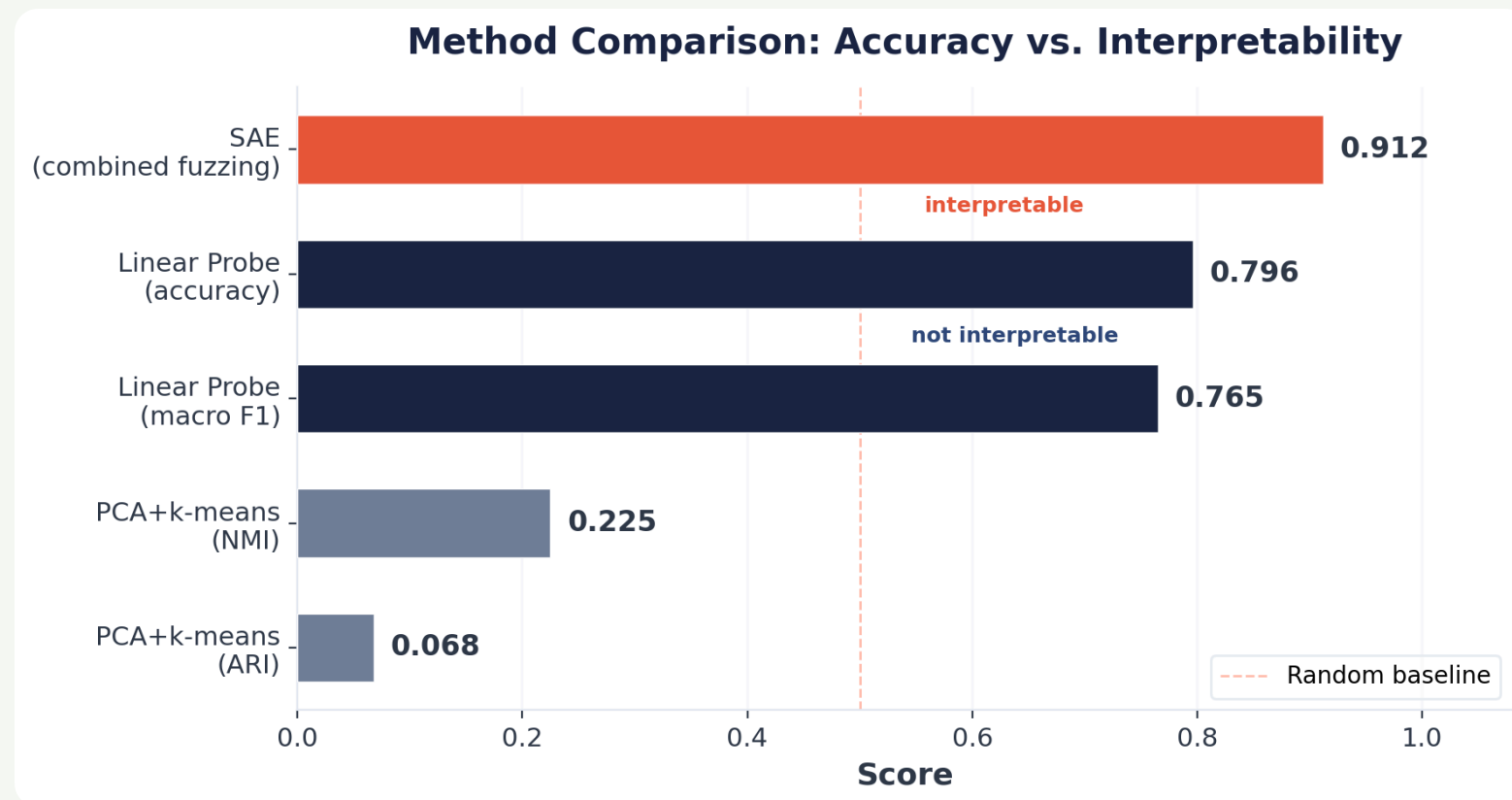
- Layers 8/16: low MSE but *pre-decision* representations
- **Layer 20**: best alive %, lowest dead %, MSE < 1.0
- Layers 32+: MoE expert routing → 500x+ higher MSE

SAE Feature Health (Layer 20, Full Dataset)

Metric	Value
Total features	10,752
Alive features (>0.1% firing)	81.2% [80.6, 81.8]
Dead features (0% firing)	0.19% [0.13, 0.27]
LO (active features per input)	668 ($\approx$ 4.1% density)
Reconstruction MSE	0.574

The SAE efficiently uses its capacity: sparse encoding with high feature utilization.

Baselines: Why Not Just a Probe?



- Linear probe confirms tool identity is *linearly encoded* (79.6% across 32 classes) -- but provides *no interpretability*
- PCA + k-means fails entirely -- tool signal is not dominant variance
- The SAE bridges this gap: **interpretable** *and* **causally testable** features

Token-Level Fuzzing Evaluation

Autointerp (EleutherAI) is the standard recipe for labelling SAE features at scale: an LLM proposes a label from top-activating contexts, a second LLM judges whether the label predicts activation. We tighten the test.

Standard evaluation: "Does this label predict which *prompts* activate the feature?"

Our evaluation: "Does this label predict which *tokens* activate the feature?"

Protocol:

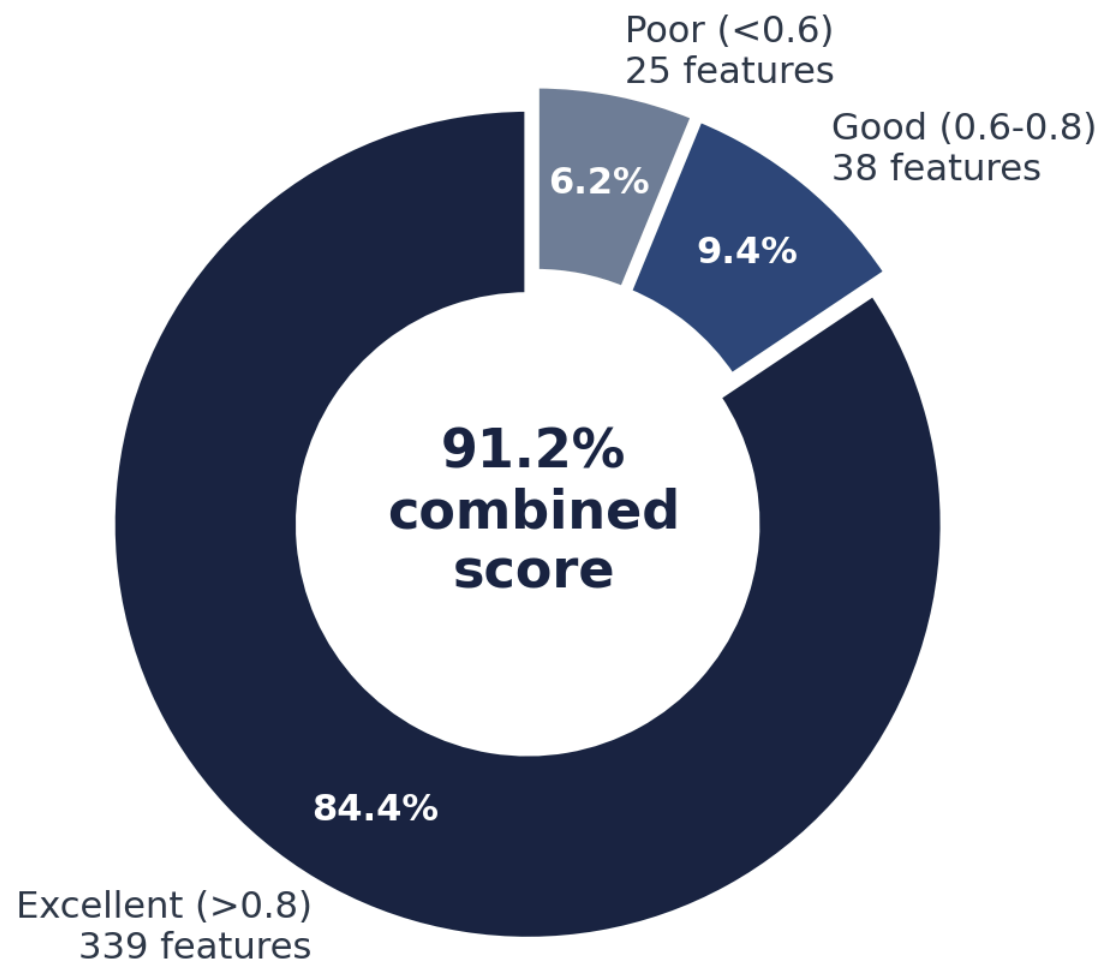
1. Extract per-token activations across entire prompt
2. Highlight top-K tokens in user request span
3. A/B test: LLM judge picks which highlighted text matches the label
4. Randomized order to prevent position bias

Combined score = $0.7 \cdot \text{acc}_{\text{token}} + 0.3 \cdot \text{acc}_{\text{prompt}}$

Token-level gets higher weight because it tests the *actual mechanism*.

Fuzzing Results: Features Are Interpretable

Feature Label Quality (Token-Level Fuzzing)



PART II

The Causality Test

From Correlation to Causal Evidence

Feature Ablation: Experimental Design

Question: Are contrastive features *causally necessary* for tool selection, or merely correlated?

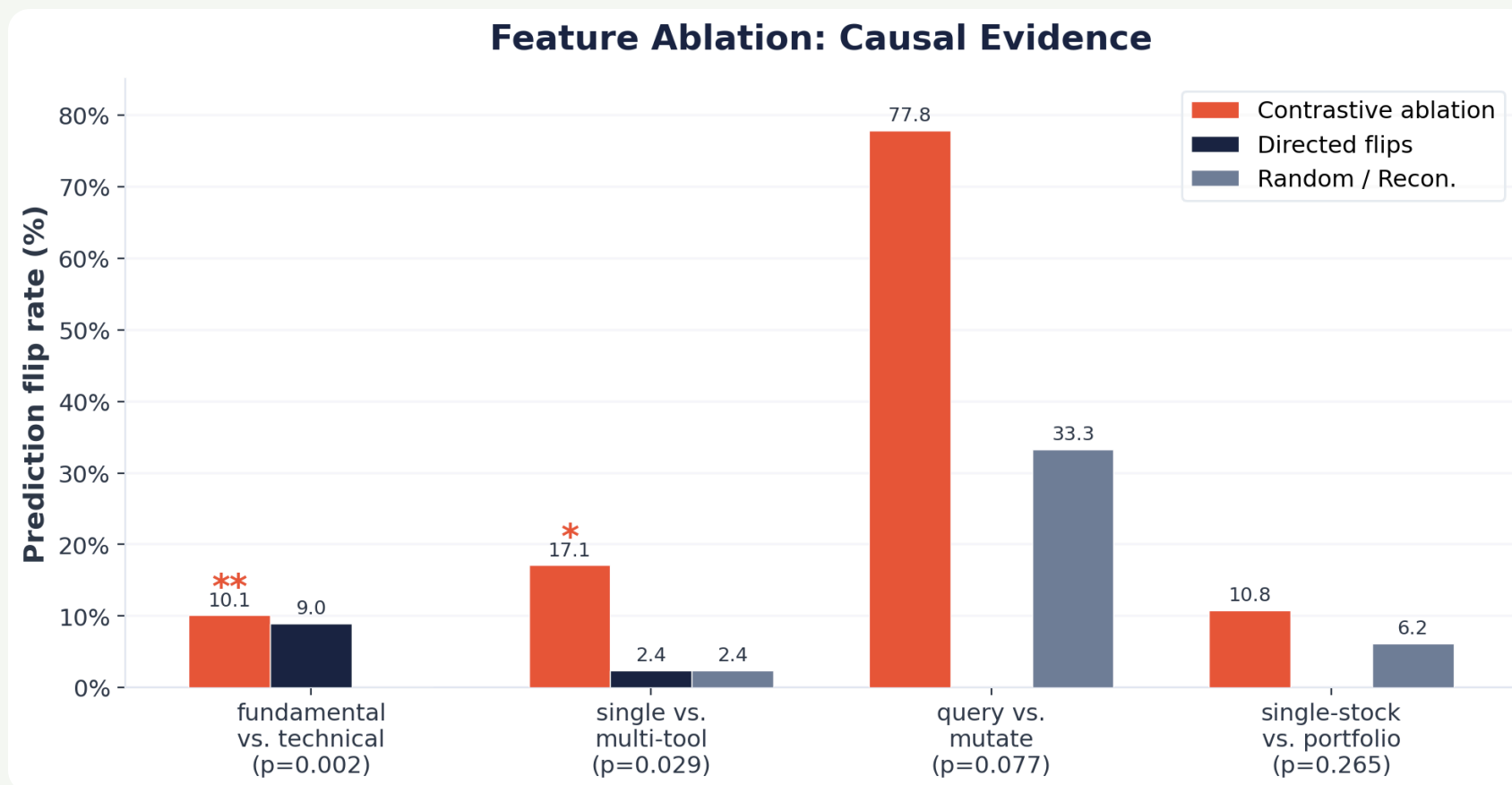
Method:

1. Intercept residual stream at layer 20
2. Encode through trained SAE
3. **Zero out top-10 contrastive features**
4. Decode back into residual stream
5. Measure: does the model's tool prediction *flip*?

Controls:

- **Random ablation:** zero 10 random non-contrastive features
- **Reconstruction-only:** SAE encode → decode with *no* features zeroed (measures round-trip distortion)

Ablation Results: Causal Evidence



Aggregate (23 types): 16.1% contrastive vs. 13.0% reconstruction-only

Why This Matters: Interpreting the Ablation

Fundamental vs. Technical Analysis ($p = 0.002$):

- Zeroing 10 contrastive features flips **10.1%** of predictions
- 9.0% flip *toward the contrast tool* (directed change)
- Random ablation: **0%** flips. Reconstruction-only: **0%** flips
- These features are *causally necessary* for this distinction

Critical control insight:

Random ablation flip rate equals SAE round-trip distortion across all 23 contrast types. Removing 10 random features from ~668 active adds no disruption beyond the encode-decode cycle itself.

This validates the design: ablation effects are due to *specific features*, not general signal degradation.

PART III

Using Kiji Inspector

FROM RESEARCH TO PRODUCTION

The Kiji Inspector



01

MODEL AGNOSTIC

Framework to train custom Sparse Autoencoders — works on **any open-source LLM**, not just the ones we tested.

02

END-TO-END TRAINING PIPELINE

No custom training setup needed. The project **generates the contrastive datasets** for you and runs the full extract → train → analyse loop.

03

PRODUCTION INFERENCE SUPPORT

vLLM patches, single-endpoint Docker containers, and PyPI packages — ready to drop into a live serving stack.

04

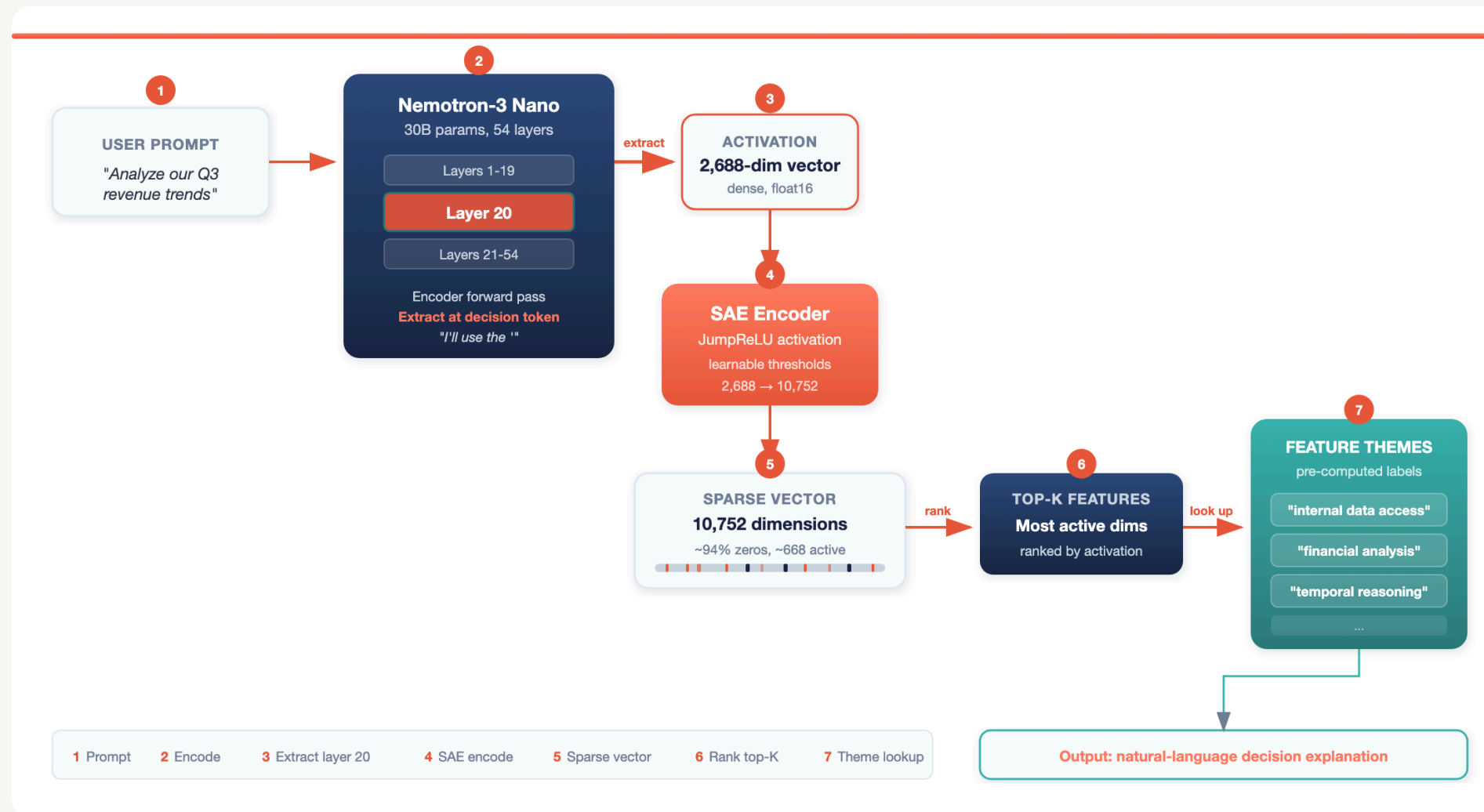
FULLY OPEN SOURCE

Pre-trained SAE models distributed via **Hugging Face, Docker Hub, and PyPI** — reuse without retraining.

Supported Language Models

Huggingface Model	Link	Kiji Inspector Sparse Autoencoder	Supported Layers
google/gemma-3-27b-it	model	575-lab/kiji-inspector-google-gemma-3-27b-it	10 20 31 41 50 58
google/gemma-4-26B-A4B-it	model	575-lab/kiji-inspector-google-gemma-4-26B-A4B-it	11 14 17 20 23
google/gemma-4-E4B-it	model	575-lab/kiji-inspector-google-gemma-4-E4B-it	11 17 23 29 35
google/gemma-4-E2B-it	model	575-lab/kiji-inspector-google-gemma-4-E2B-it	13 14 15 18 19 20 23 24 25
nvidia/Nemotron-3-Super-120B-A12B-FP8	model	575-lab/kiji-inspector-NVIDIA-Nemotron-3-Super-120B-A12B-FP8	15 27 45 57 69 81
nvidia/Nemotron-3-Super-120B-A12B-BF16	model	575-lab/kiji-inspector-NVIDIA-Nemotron-3-Super-120B-A12B-BF16	15 27 45 57 69 81
nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16	model	575-lab/kiji-inspector-NVIDIA-Nemotron-3-Nano-30B-A3B-BF16	8 17 20 26 35 44
nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-FP8	model	575-lab/kiji-inspector-NVIDIA-Nemotron-3-Nano-30B-A3B-FP8	8 17 20 26 35 44
nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-NVFP4	model	575-lab/kiji-inspector-NVIDIA-Nemotron-3-Nano-30B-A3B-NVFP4	8 17 20 26 35 44

Our Inference Setup



Seven steps from raw prompts to human-readable decision explanations.

Hands-on: Extract a Layer's Activations

The Kiji Inspector builds on standard PyTorch primitives. Here's the extraction step in raw `transformers` — the same mechanism, no Kiji magic, on any HuggingFace causal LM.

```
pip install -U -q kiji-inspector transformers
```

Setup — Imports & Config

```
import torch
from transformers import AutoModelForCausalLM, AutoProcessor

LAYER_INDEX = 8
MODEL_ID = "google/gemma-4-E4B-it"
PROMPT = "My dishwasher is smelly, what is the first element I should review?"
```

Pick the **layer** to study and the **model** to study it on. Everything else flows from these three constants.

Load Model & Processor

```
processor = AutoProcessor.from_pretrained(MODEL_ID)
model = AutoModelForCausalLM.from_pretrained(
    MODEL_ID,
    torch_dtype=torch.bfloat16,
    device_map="auto",
)
model.eval()
```

`bfloat16` halves memory; `device_map="auto"` lets HF Accelerate place layers across GPUs. `.eval()` disables dropout — we want deterministic activations.

Format the Prompt

```
messages = [{"role": "user", "content": [{"type": "text", "text": PROMPT}]]
prompt = processor.apply_chat_template(
    messages, tokenize=False, add_generation_prompt=True,
)
inputs = processor(text=prompt, return_tensors="pt")
inputs = {k: v.to(next(model.parameters()).device) for k, v in inputs.items()}
```

`add_generation_prompt=True` appends the assistant turn header — the model is now poised to *answer*. That final position is the **decision token** we just defined.

Capture Activations with a Forward Hook

```
captured = {}

def hook(_module, _inputs, output):
    hidden = output[0] if isinstance(output, tuple) else output
    captured["tensor"] = hidden.detach().cpu()

layer = model.model.language_model.layers[LAYER_INDEX]
handle = layer.register_forward_hook(hook)
```

PyTorch's `register_forward_hook` intercepts the layer's output mid-forward-pass. We `.detach()` to drop the autograd graph and `.cpu()` so we don't pin GPU memory.

Run Inference, Retrieve the Tensor

```
try:
    with torch.inference_mode():
        model(**inputs)
finally:
    handle.remove()

hidden_state = captured["tensor"]
```

`inference_mode()` is faster than `no_grad()` — it also skips view-tracking. The `try / finally` guarantees the hook is removed even on error. `hidden_state` is ready to feed into a trained SAE.

Now: Ask the Kiji Inspector What Fired

```
from kiji_inspector import SAE

sae, feature_descriptions = SAE.from_pretrained(
    base_model="google/gemma-4-E4B-it",
    layer=8,
)

# Activation for the first sequence, last token
last_token_act = hidden_state[0, -1, :]

# Describe the top features activating on this token
sae.describe(last_token_act, feature_descriptions)
```

`SAE.from_pretrained` pulls a layer-matched SAE *and* its labelled feature dictionary from the Hub. `describe()` returns the highest-activating features and their human-readable labels — the decision token, explained.

Output: Top Features on the Token

```
[(2341, 'unknown', 7.57),  
 (14641,  
  {'label': 'Dishwasher Gasket Issues',  
   'description': 'Detects descriptions of dishwashers leaking from '  
                   'the door or bottom, often mentioning a worn, torn, '  
                   'or cracked door gasket.',  
   'confidence': 'high',  
   'mean_activation': 7.66,  
   'max_activation': 8.63,  
   'frac_nonzero': 1.0,  
   'top_examples': ['My 18-yr-old dishwasher is leaking from the door...',  
                    ...],  
   'bottom_examples': ['Find general best practices for API versioning...',  
                       ...]}},  
 7.19),  
 ...]
```

A labelled feature (**#14641**) semantically matches the prompt — with full provenance (confidence, activation stats, witnessing contexts). Feature **#2341** fired too but autointerp hasn't named it.

Demo Time

KIJI INSPECTOR, LIVE



github.com/dataiku/kiji-inspector

End-to-End Demo Application

How Your AI Advisor Thinks

See the reasoning behind every repair recommendation — what data the AI gathered, what factors it weighed, and why it reached each conclusion.

Showing live results from this demo run.

► How to read this page

SELECT A PROBLEM



Dishwasher leaking water from the bottom

Bosch 500 Series dishwasher · 3 years old

FIX SOON **MODERATE** **\$13-\$90 DIY**



Garbage disposal hums but won't spin

InSinkErator Badger 5 · 2 years old

CAN WAIT **EASY** **\$0-\$8 DIY**



Water heater making loud popping and rumbling sounds

Rheem 50-gallon gas water heater · 9 years old

ACT NOW **DIFFICULT** **\$15-\$649 DIY**

APPLIANCE

Rheem 50-gallon gas water heater

AGE

9 years

SYMPTOMS

Loud popping when heating up, especially in the morning. Hot water takes longer to reach faucets. Slight rust tinge in the first few seconds of hot water.



Step 1: Repair Manual Lookup

Possible causes: Sediment buildup on tank bottom, Anode rod depleted, Scale buildup on burner assembly reducing heat transfer efficiency, Possible tank corrosion if popping is severe and rust is present in water

DIY difficulty: Moderate to Difficult

Safety: GAS APPLIANCE: Risk of scalding burns from hot water and gas leaks if connections are disturbed. Turn off gas supply valve

WHAT THE AI NOTICED

Feature #3062 *strong*

Old Machine Troubleshooting *strong*

Feature #4866 *strong*

Feature #3776 *moderate*

Interactive system surfacing SAE-derived explanations alongside agent output -- translating internal feature activations into natural-language rationales.

Ideas for Your Use Cases of Kiji Inspector



01

DEBUG AGENT MISBEHAVIOUR

When the agent picks the *wrong* tool, read off the firing features to see what the model actually keyed on — not the story its chain-of-thought tells.

02

AUDIT DECISIONS FOR COMPLIANCE

Catch agents routing on signals they shouldn't — sensitive attributes, leaked context, prompt-injected instructions. Evidence beyond input/output logs.

03

VALIDATE PROMPT ENGINEERING

Confirm a prompt edit actually shifted the *decision-relevant* features — not just the output text on a handful of examples.

04

MONITOR PRODUCTION DRIFT

Track feature-activation distributions over time. Alert when the agent's internal rationale moves, even if the tool-choice metrics still look fine.

Key Takeaways

01

SAEs discover interpretable decision features without supervision — 91.2% fuzzing accuracy

02

Token-level fuzzing catches labels *“right for the wrong reasons”* — a stricter test than prompt-level evaluation

03

Causal evidence via ablation: specific features are necessary for specific tool-selection decisions ($p = 0.002$)

04

The reconstruction-only baseline is essential — it separates genuine causal effects from SAE distortion artifacts

Limitations and Future Directions

CURRENT LIMITATIONS

- Training is compute-intensive
(235B generation model + 30B subject model)
- Access to model is required
- Synthetic contrastive pairs may miss real-world decision factors

FUTURE WORK

- Support for more open-source models like Qwen
- Multi-layer / circuit-level analysis
- Cross-model transfer (do tool-selection circuits generalize?)

Thank You

Open source: github.com/dataiku/kiji-inspector



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